

Optical Frequency Combs: Principles and Operation

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Abstract

An optical frequency comb is a laser source whose spectrum consists of a large number of discrete, equally spaced lines spanning a broad optical bandwidth. Generated by a mode-locked femtosecond laser, the comb provides a direct, phase-coherent link between radio-frequency standards and optical frequencies. This document explains the physical origin of the comb, the two degrees of freedom that must be stabilised (f_{ceo} and f_{rep}), the f -to- $2f$ self-referencing technique used to detect the carrier-envelope offset, and the procedure for phase-locking an external continuous-wave laser to a comb tooth via beatnote detection. The conditions under which a beatnote lock fails — including the pull-out condition for a PLL tracking a rapidly shifting beatnote — are discussed, along with digital and hardware strategies for improving lock robustness.

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1 Physical Origin of the Frequency Comb

1.1 Mode-Locked Lasers

A mode-locked laser produces a periodic train of ultrashort optical pulses by forcing many longitudinal cavity modes to oscillate simultaneously with a fixed phase relationship. The round-trip time of the cavity is $T_{\text{rep}} = 2\frac{L}{v_g}$, where L is the cavity length and v_g is the group velocity of the pulse envelope. Each pulse is separated from the next by exactly T_{rep} , giving a pulse repetition frequency

$$f_{\text{rep}} = \frac{1}{T_{\text{rep}}}$$

In practice, f_{rep} lies in the range 100 MHz to 10 GHz depending on the cavity length. The pulse duration is typically 10–100 fs, making these lasers extremely broadband: a 30 fs pulse centred at 800 nm has a transform-limited spectral width of roughly 30 THz.

1.2 From Pulse Train to Frequency Comb

The Fourier transform of a perfectly periodic train of identical pulses is a **comb** of Dirac delta functions, each separated by f_{rep} . Real mode-locked lasers introduce one additional complication: the group velocity and phase velocity inside the cavity are not equal due to dispersion. As a result, the carrier wave of each pulse is slightly phase-shifted relative to the envelope of the previous pulse by an amount $\Delta\varphi_{\text{ceo}}$ per round trip. This carrier-envelope phase slip accumulates from pulse to pulse and shifts the entire comb spectrum away from zero by the **carrier-envelope offset frequency**:

$$f_{\text{ceo}} = \frac{\Delta\varphi_{\text{ceo}}}{2\pi} \cdot f_{\text{rep}}, \quad f_{\text{ceo}} \in [0, f_{\text{rep}})$$

The frequency of the n -th comb tooth is therefore

$$f_n = f_{\text{ceo}} + n \cdot f_{\text{rep}}$$

where n is a large positive integer — typically 3×10^5 to 10^6 for teeth in the visible or near-infrared. Because f_{rep} and f_{ceo} are both radio frequencies (below 1 GHz), they can be measured directly with fast electronics and stabilised to an absolute reference with sub-hertz precision. Every comb tooth is then phase-coherently anchored to that reference, realising an **optical clockwork** that connects the microwave and optical domains.

Quantity	Physical meaning	Typical value
f_{rep}	Pulse repetition rate; tooth spacing	250 MHz – 1 GHz
f_{ceo}	Carrier-envelope offset; comb shift from zero	$0 - f_{\text{rep}}$
n	Tooth index	$3 \times 10^5 - 10^6$
f_n	Absolute frequency of the n -th tooth	100 THz – 1 PHz

Table 1: The four quantities in the comb equation.

2 Stabilisation of the Two Degrees of Freedom

A fully stabilised comb requires two independent servo loops: one to fix f_{rep} and one to fix f_{ceo} . Once both are locked, every tooth f_n is absolutely determined.

2.1 Stabilisation of f_{rep}

A fast photodetector illuminated by the pulse train produces a photocurrent at f_{rep} and its harmonics $2f_{\text{rep}}, 3f_{\text{rep}}, \dots$. The fundamental tone at f_{rep} is extracted with a bandpass filter and mixed against a radio-frequency reference oscillator — typically a GPS-disciplined quartz oscillator or hydrogen maser — to generate a phase error signal. This error signal is fed back to the cavity length via a piezoelectric mirror mount or an intracavity electro-optic modulator. Once the loop is closed, f_{rep} inherits the long-term absolute accuracy and short-term phase noise of the microwave reference.

The fractional frequency instability achievable is limited by the quality of the microwave reference: $< 10^{-13}$ at 1 s for a GPS-disciplined oscillator, $< 10^{-15}$ at 1 s for a hydrogen maser. Multiplied up to optical frequencies by the factor $n \approx 5 \times 10^5$, the absolute frequency uncertainty of a comb tooth is at the millihertz level — negligible for virtually all optical spectroscopy applications.

2.2 Detection of f_{ceo} : the f -to- $2f$ Interferometer

The carrier-envelope offset f_{ceo} cannot be detected by a simple photodetector because no electronic device can follow optical oscillations directly. Instead, f_{ceo} is measured through **self-referencing**, which requires the comb to span a full optical octave (a factor of 2 in frequency). If the comb runs from f_{low} to $f_{\text{high}} = 2f_{\text{low}}$, then two teeth at index n and $2n$ exist:

$$f_n = f_{\text{ceo}} + n \cdot f_{\text{rep}}$$

$$f_{\{2n\}} = f_{\text{ceo}} + 2n \cdot f_{\text{rep}}$$

Frequency-doubling f_n in a nonlinear crystal gives $2f_n = 2f_{\text{ceo}} + 2n \cdot f_{\text{rep}}$. Beating this against $f_{\{2n\}}$ on a photodetector yields

$$2f_n - f_{\{2n\}} = (2f_{\text{ceo}} + 2n \cdot f_{\text{rep}}) - (f_{\text{ceo}} + 2n \cdot f_{\text{rep}}) = f_{\text{ceo}}$$

The n -dependence cancels exactly, leaving a pure radio-frequency signal at f_{ceo} that is detectable with standard electronics. This is the **f -to- $2f$ interferometer**.

In practice, few mode-locked oscillators are natively octave-spanning. The comb spectrum is first broadened to a full octave using a nonlinear medium — most commonly a few centimetres of photonic crystal fibre (PCF), which has a carefully engineered dispersion profile near the comb's centre wavelength. A dichroic beamsplitter then separates the low- and high-frequency wings; the low-frequency wing is frequency-doubled and recombined with the high-frequency wing for heterodyne detection.

2.3 Stabilisation of f_{ceo}

The f_{ceo} signal detected by the f -to- $2f$ interferometer is mixed against a reference frequency and the error signal is fed back to the pump power of the mode-locked oscillator. Changing the pump power modifies the nonlinear phase inside the gain medium, which shifts $\Delta\varphi_{\text{ceo}}$ per round trip and hence f_{ceo} . The bandwidth of this pump-power servo is typically a few kilohertz, sufficient to suppress low-frequency f_{ceo} fluctuations caused by pump laser intensity noise.

With both loops closed, the comb equation becomes a phase-coherent equality: every tooth f_n is an exact multiple-plus-offset of the microwave reference, with no free parameters.

3 Locking a CW Laser to the Comb

3.1 Beatnote Detection

To transfer the comb's frequency accuracy to a continuous-wave (CW) laser at frequency f_{laser} , a small fraction of each beam is combined on a fast photodetector — the **beatnote detector**. The photocurrent contains a component at the difference frequency between f_{laser} and the nearest comb tooth f_n :

$$f_{\text{beat}} = |f_{\text{laser}} - f_n| = |f_{\text{laser}} - f_{\text{ceo}} - n \cdot f_{\text{rep}}|$$

The tooth index n is determined from a coarse frequency measurement of the CW laser (e.g. a wavemeter) using $n = \text{round}\left(\frac{f_{\text{laser}} - f_{\text{ceo}}}{f_{\text{rep}}}\right)$. The beatnote f_{beat} is a radio-frequency signal in the range $\left[0, \frac{f_{\text{rep}}}{2}\right]$ and appears as a peak on an RF spectrum analyser.

3.2 Phase-Locked Loop Architecture

The beatnote is phase-locked to a stable reference frequency f_{ref} (often a few tens of megahertz) by an electronic servo loop. The phase error $\varphi_{\text{err}} = \varphi_{\text{beat}} - \varphi_{\text{ref}}$ drives an actuator on the CW laser — most commonly:

- **Acousto-optic modulator (AOM):** shifts the laser frequency by $\pm$ several tens of MHz with a servo bandwidth of several megahertz. Used for fast corrections.
- **Electro-optic modulator (EOM):** provides even higher bandwidth (tens of MHz) but smaller frequency range.

- **Laser piezo or grating:** provides a large frequency range but slow response (bandwidth < 10 kHz). Used for slow drift correction.

A common architecture combines an AOM for fast corrections with a piezo for slow corrections, paralleling the two-actuator approach used for the 689 nm laser in the main locking system.

When the beatnote lock is closed and both f_{rep} and f_{ceo} are stabilised, the CW laser frequency is absolutely determined:

$$f_{\text{laser}} = f_{\text{ceo}} + n \cdot f_{\text{rep}} + f_{\text{ref}}$$

The uncertainty is set by the uncertainties of f_{ceo} , f_{rep} , and f_{ref} added in quadrature — at the $10^{\{-15\}}$ fractional level for a GPS-referenced system.

3.3 Identifying the Correct Tooth

Because the beatnote is symmetric about zero, f_{beat} is the same whether $f_{\text{laser}} = f_n + f_{\text{beat}}$ or $f_{\text{laser}} = f_n - f_{\text{beat}}$. The sign ambiguity is resolved by slightly changing a known comb parameter (e.g. f_{rep}) and observing which direction f_{beat} shifts. Once the sign is established, the servo sign is set and the lock is engaged.

4 Beatnote Lock Failure Modes

4.1 PLL Pull-Out: the Fundamental Bandwidth Limit

A phase-locked loop can track a frequency-modulated input only up to a maximum rate of frequency change. For a second-order PLL with natural frequency $\omega_n = 2\pi B$ (where B is the loop bandwidth), the pull-out condition is

$$\dot{f}_{\text{beat}} > 2\pi B^2$$

If the beatnote shifts faster than this rate, the phase error grows without bound and the loop loses lock. This is the primary failure mode when the CW laser or the comb experiences a rapid perturbation. A loop with bandwidth $B = 1$ MHz can track beatnote slew rates up to roughly $6 \times 10^{\{12\}}$ Hz/s; a software-based loop with $B = 1$ kHz can track only 6×10^6 Hz/s — six orders of magnitude less.

4.2 Common Causes of Rapid Beatnote Shifts

Acoustic perturbations: vibrations that couple into the CW laser cavity (e.g. via the optical table or the fibre coupling to the comb) cause fast frequency fluctuations. Acoustic isolation and rigid fibre mounts reduce this.

Fibre phase noise: if the comb light is delivered to the beatnote detector via a long optical fibre, acoustic and thermal noise on the fibre add phase noise to the comb teeth, appearing as a fluctuating f_{beat} . Fibre noise cancellation (sending a reflected beam back and locking the round-trip phase) eliminates this.

f_{ceo} or f_{rep} transients: a glitch in the pump laser or cavity can momentarily shift f_{ceo} by a large amount, pulling f_{beat} out of the servo capture range. A robust f_{ceo} loop with sufficient bandwidth prevents this.

CW laser mode hops: a sudden jump in the CW laser frequency by a cavity free spectral range (several GHz) moves f_{beat} to a completely different tooth. This is not a PLL failure but a physical frequency discontinuity; the lock must be re-acquired from scratch.

4.3 Strategies for Improving Lock Robustness

Strategy	Mechanism	Bandwidth gain
FPGA-based PLL	Replaces software loop; reduces round-trip latency to $< 1 \mu\text{s}$	$\times 10^3$
AOM fast actuator	High-bandwidth frequency shifter; no mechanical resonances	Up to $\sim 5 \text{ MHz}$
Feedforward of f_{rep} , f_{ceo}	Cancels known comb fluctuations before they reach the beatnote servo	Reduces residual $\dot{f}_{\text{beat}}$
Fibre noise cancellation	Locks the fibre path length; removes fibre-induced phase noise	Removes a noise floor
Acoustic isolation	Reduces the source of fast perturbations	Problem-dependent
Wider capture range PLL	Frequency-to-digital converter detects f_{beat} over $> \text{one octave}$	Extends re-capture after transients

Table 2: Strategies for improving beatnote lock robustness.

Feedforward deserves special mention for digital implementations. If the error signals from the f_{rep} and f_{ceo} stabilisation loops are available as digital monitor outputs, they encode the instantaneous fluctuations of every comb tooth. These can be combined and applied as a feedforward correction to the CW laser actuator, reducing the residual beatnote motion seen by the servo loop. This relaxes the bandwidth requirement and can be implemented entirely in software if the comb electronics provide suitable outputs.

5 Summary

The frequency comb is characterised by two radio-frequency parameters — f_{rep} and f_{ceo} — that together specify the position of every tooth in the optical spectrum via $f_n = f_{\text{ceo}} + n \cdot f_{\text{rep}}$. Stabilising both parameters to a microwave reference transfers its accuracy to every tooth. A CW laser is locked to the comb by phase-locking the beatnote $f_{\text{beat}} = |f_{\text{laser}} - f_n|$ to a reference frequency, yielding an absolutely known optical frequency $f_{\text{laser}} = f_{\text{ceo}} + n \cdot f_{\text{rep}} + f_{\text{ref}}$. The primary failure mode is a rapidly shifting beatnote exceeding the PLL pull-out limit $\dot{f}_{\text{beat}} = 2\pi B^2$; the remedy is higher servo bandwidth, faster actuators, or feedforward cancellation of the comb fluctuations.